

Bloodrose — Render Pipeline Spec

Governed Presentation Architecture

Project	Bloodrose
Artifact	Render Pipeline Spec
Version	v0.1
Classification	Governed Presentation Architecture
Date	April 21, 2026
Status	Active
Depends On	Bloodrose Canonical Spec v0.1, Modular Architecture v0.1, Canonical Event Table v0.1, Story Forge Integration Layer v0.1

Audience: Render engineers, visual artists, audio designers, UI/UX designers, system architects.

2. Render Philosophy

The Render Pipeline exists to make state visible. Every visual element, every line of narrative text, every sound cue must be traceable to a canonical state value or event. Atmosphere is permitted — encouraged — but never at the expense of mechanical legibility. The player should always be able to read the game state through the presentation. A crimson screen means corruption is high. A golden glow means redemption is within reach. A wilted petal on the ground means a Guilt collectible is present. There is no ambiguity between what the player sees and what the system knows.

"Atmosphere may be expressive. Logic must remain explicit."

This is the render pipeline's governing principle. The crimson glow of high corruption, the quiet gold of rising redemption, the pulse of a terminal event — these are governed visual expressions of state, not decorative flourishes. Nothing on screen is arbitrary. Every color shift, every particle trail, every ambient sound change corresponds to a defined state range or event ID documented in this specification. An artist may choose the shape of a thorn, but the thorn's presence, its color intensity, and its animation behavior are dictated by state.

The Render Pipeline is a pure consumer. It reads state from the State Manager, entity positions from the Entity Layer, and narrative dispatches from the Event System. It never writes back. This separation is enforced by the Modular Architecture (Section 4.7) and validated by the Drift Detection Harness — DFT-012: Render State Mutation is a **CRITICAL** violation. If the Render Pipeline ever modifies corruption, redemption, score, or any canonical state value, the build fails. No exceptions.

The pipeline is designed for browser-native rendering (HTML5 Canvas or DOM) with no external plugins. Visual fidelity serves mood and legibility, not technical showcase. The aesthetic is crimson, black, moonlit, gothic, surreal romantic horror — beauty woven with danger, heartbreak made physical. Lady Isolde's heart-garden should feel like a place that was once beautiful and still is, if you don't look too closely at the thorns. The render pipeline's task is to ensure the player always knows exactly how close those thorns are.

3. Pipeline Architecture

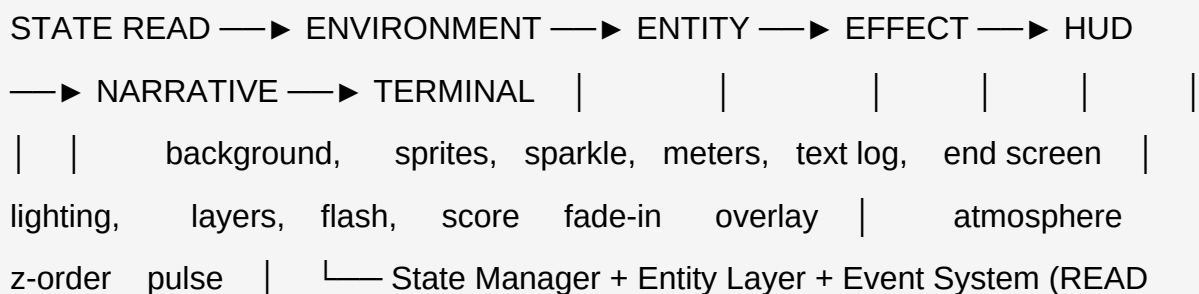
3.1 Pipeline Stages

The render pipeline executes the following stages in strict sequence every frame. No stage may be skipped or reordered. Each stage reads inputs established during Stage 1 and produces visual output to a single composited frame buffer.

1. **Stage 1 — STATE READ:** Read current values from State Manager (corruption, redemption, score, game_over, ending_type). Read entity positions and states from Entity Layer. Receive any pending narrative dispatches from Event System. This is the *only* data acquisition point — no mid-frame reads.

2. **Stage 2 — ENVIRONMENT RENDER:** Draw the Heart-Garden background. Apply state-driven environmental effects: corruption level affects darkness, thorn density, and color saturation; redemption level affects ambient light and petal glow intensity.
3. **Stage 3 — ENTITY RENDER:** Draw all game entities in layer order: thorn zones (bottom), petals (middle), altar (middle), player (top). Each entity type has a defined visual representation and state-driven visual variants as specified in Section 4.3.
4. **Stage 4 — EFFECT RENDER:** Draw transient visual effects triggered by events (collection sparkle, thorn damage flash, threshold transition pulse, altar activation glow). Effects are time-limited and do not persist beyond their defined animation duration.
5. **Stage 5 — HUD RENDER:** Draw the heads-up display: corruption meter, redemption meter, score readout. HUD elements update smoothly via animated interpolation but must converge to true state values within 500ms.
6. **Stage 6 — NARRATIVE RENDER:** Update the rolling narrative log with any new dispatches from the Event System. Narrative text fades in, remains visible for a configured duration, then scrolls upward as new entries arrive.
7. **Stage 7 — TERMINAL RENDER:** If `game_over` is true, render the appropriate end screen (*Consumed by Thorns*, *Mercy Dominant*, or *Bound but Unbroken*) with full-screen overlay, ending narrative text, and restart prompt.

3.2 Pipeline Diagram



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GOVERNANCE NOTE

The arrow from Stage 1 flows in one direction only. No stage writes back to the State Manager, Entity Layer, or Event System. This is architecturally enforced (Modular Architecture §4.7) and validated at runtime (Drift Detection DFT-012).

4. Visual Specification

4.1 Color Palette

The following table defines the canonical color system for Bloodrose. All render code must reference these values. Deviations require a formal amendment to this document (see Section 15).

Color Name	Hex Value	Usage	State Association
Crimson Deep	#8B0000	Primary accent, thorn highlights, corruption meter fill	Corruption
Blood Red	#CC0000	Petal glow (Guilt, Rage), damage flash	High corruption events
Rose Pink	#FF6B8A	Petal glow (Desire), ambient particles	Desire / mixed state
Gold Warm	#D4A574	Petal glow (Mercy), altar glow, redemption meter fill	Redemption
Moonlight Silver	#C0C0C0	Ambient light at neutral state, text highlights	Neutral

Color Name	Hex Value	Usage	State Association
Midnight Black	#0A0A0A	Primary background, deep shadows	Base environment
Charcoal	#1A1A1A	Secondary background, thorn zone base	Hazard areas
Ash Gray	#333333	Inactive UI elements, faded text	Low-priority display
Thorn Green	#2D4A2D	Subtle thorn vine coloring	Organic hazard elements
Ethereal White	#F0E6D3	Narrative text, altar radiance at full redemption	Clarity / resolution
Corruption Violet	#4A0040	Environmental tint at high corruption	Corruption atmosphere
Garden Fog	#1A1520	Environmental fog, depth layering	Atmospheric depth

4.2 State-Driven Color Mapping

The environment's color temperature shifts in real time based on the corruption and redemption state axes. Both axes render simultaneously — their visual effects compose additively. A player at corruption 60 / redemption 70 sees the red-violet tint *and* the warm glow. The visual tension *is* the narrative.

Corruption Axis

State Range	Environment al Tint	Ambient Light Level	Particle Color	Mood Keyword
corruption 0–24	No tint overlay	Normal ambient	Neutral particles	Calm, fragile
corruption 25–49	Slight reddish tint (#1A0505 overlay at 10% opacity)	Ambient dims 10%	Neutral	Tension rising
corruption 50–74	Red-violet tint (#2A0515	Ambient dims 25%	Shift toward red	Danger, closing in

State Range	Environmental Tint	Ambient Light Level	Particle Color	Mood Keyword
	overlay at 25% opacity)			
corruption 75–99	Deep violet tint (#3A0525 overlay at 40% opacity)	Ambient dims 40%; screen edge vignette	Aggressive red	Critical, near collapse
corruption 100	Full blackout transition over 2 seconds	Fades to zero	Crimson pulse before fade	Consumed

Redemption Axis

State Range	Environmental Tint	Ambient Light Level	Particle Color	Mood Keyword
redemption 0–29	No redemption visual effect	No change	None	Unawakened
redemption 30–64	Subtle warm glow around player (gold at 10% intensity)	No global change	Occasional gold particle	Stirring, not yet enough
redemption 65+	Warm ambient increase (15%)	Altar emits visible glow	Gold particles around altar	Path open, possibility alive

IMPORTANT — Dual-Axis Composition

Both axes render simultaneously. Their visual effects compose additively. A player at corruption 60 / redemption 70 sees the red-violet environment tint *AND* the warm golden glow around the altar. This visual tension — crimson and gold fighting for dominance on screen — is a deliberate aesthetic expression of the player's state. Do not suppress either axis.

4.3 Entity Visual Specifications

Player (Lady Isolde)

Property	Specification
Shape / Form	Humanoid sprite or distinct glowing silhouette with flowing form suggesting a cloaked figure
Size	1.0 unit (reference scale)
Base Color	Pale silhouette with crimson/gold edge glow depending on dominant state
State Variants	Edge glow shifts from gold (redemption dominant) to crimson (corruption dominant). At corruption ≥ 75 , thorny tendrils visually grow from sprite edges.
Animation	Idle breathing animation (subtle scale pulse). Movement trailing particles matching dominant state color.
Layer	Top (z-index highest among game entities)

Guilt Petal

Property	Specification
Shape / Form	Wilted rose petal, curled inward
Size	0.4 units
Base Color	Deep red (#CC0000) with dark edges
Animation	Slow pulse (opacity 0.7–1.0, 2-second cycle); faint smoke wisps
Collection Effect	Burns inward (shrink + red flash) over 300ms
Layer	Middle

Desire Petal

Property	Specification
Shape / Form	Half-open rose petal, lush and ambiguous
Size	0.4 units
Base Color	Rose pink (#FF6B8A) with warm gold shimmer
Animation	Gentle sway (position oscillation ± 2 px, 3-second cycle); dual-color particle trail (pink and gold)
Collection Effect	Splits into two dissolving fragments (one red, one gold) over 400ms
Layer	Middle

Rage Petal

Property	Specification
Shape / Form	Jagged petal, torn and angular
Size	0.45 units (slightly larger — aggressive presence)
Base Color	Intense crimson (#FF0000) with black vein patterns
Animation	Aggressive pulse (opacity 0.8–1.0, 1-second fast cycle); ember particles
Collection Effect	Explodes outward (brief radial flash + ember burst) over 250ms
Layer	Middle

Mercy Petal

Property	Specification
Shape / Form	Full, open rose petal, soft and luminous
Size	0.4 units
Base Color	Warm gold (#D4A574) with ethereal white glow

Property	Specification
Animation	Steady gentle glow (opacity 0.85–1.0, 4-second slow cycle); drifting light motes
Collection Effect	Expands outward (brief golden halo) then absorbs into player over 500ms
Layer	Middle

Thorn Zone

Property	Specification
Shape / Form	Irregular cluster of intertwined thorny vines covering a defined rectangular area
Size	Variable (defined per instance, minimum 2×2 units)
Base Color	Dark thorn green (#2D4A2D) with charcoal shadows
State Variants	At corruption ≥ 50 , thorns appear thicker and redder. At corruption ≥ 75 , thorns subtly writhe (slow position jitter animation).
Contact Effect	When player overlaps: red pulse radiates from point of contact; screen edge flashes red briefly (100ms flash, then fade).
Layer	Bottom (below entities, above background)

Altar

Property	Specification
Shape / Form	Ornate stone altar with rose motif, centered on a clear area
Size	2.0 × 1.5 units
Base Color	Dark stone gray (#444444) with crimson rose engravings

Property	Specification
Locked State (redemption < 65)	Dark, inert. Engravings barely visible. Faint crimson outline.
Unlocked State (redemption ≥ 65)	Engravings glow warm gold. Light column rises from altar. Ambient particles orbit. Visible from across the play area.
Rejected Effect (EVT-033)	Brief crimson barrier flash when player touches while locked.
Layer	Middle (same as petals)

5. HUD Specification

5.1 Layout

The HUD occupies the top edge of the screen. All HUD elements are rendered on a semi-transparent dark background panel (#0A0A0A at 70% opacity) to ensure readability against any environment state.

- **Corruption Meter:** Left-aligned horizontal bar, fills left to right, colored by corruption level.
- **Score Display:** Centered between meters, numeric readout.
- **Redemption Meter:** Right-aligned horizontal bar, fills right to left, colored by redemption level.

5.2 Corruption Meter

Property	Specification
Width	30% of screen width
Height	16px with 1px border
Fill Color	Gradient from #8B0000 (low) to #FF0000 (high)
Background	#1A1A1A
Border	1px solid #333333
Label	"CORRUPTION" in small caps,

Property	Specification
	#C0C0C0, 10px, left-aligned above bar
Threshold Markers	Subtle tick marks at 50 and 75 on the bar
At corruption ≥ 75	Bar pulses (opacity animation 0.8-1.0, 1-second cycle)
At corruption ≥ 90	Bar pulses faster (0.5-second cycle) and border glows red

5.3 Redemption Meter

Property	Specification
Width	30% of screen width
Height	16px with 1px border
Fill Color	Gradient from #8B6914 (low) to #D4A574 (high)
Background	#1A1A1A
Border	1px solid #333333
Label	"REDEMPTION" in small caps, #C0C0C0, 10px, right-aligned above bar
Threshold Markers	Subtle tick marks at 30 and 65 on the bar
At redemption ≥ 65	Bar emits steady warm glow (gold)

5.4 Score Display

Property	Specification
Position	Centered horizontally, same vertical line as meters
Font	Consolas or serif, 14px, #C0C0C0
Format	"SCORE: [value]"
Animation	On score increase, brief white flash on the number (200ms)

6. Narrative Log Specification

6.1 Layout

Property	Specification
Position	Bottom of screen, full width
Height	Variable, showing 3–4 most recent entries; older entries scroll upward and fade
Background	Semi-transparent dark panel (#0A0A0A at 60% opacity)
Text Style	Italic, Cambria (serif), #F0E6D3, 12px
Line Spacing	1.4em

6.2 Entry Behavior

8. New entries fade in over 500ms (opacity 0 → 1).
9. Entries remain at full opacity for **8 seconds**.
10. After 8 seconds, entries begin fading (opacity 1 → 0.4 over 3 seconds) and scroll upward as new entries arrive.
11. Maximum visible entries: **4**. Oldest entry fully fades when the 5th entry arrives.
12. Terminal event narrative overrides the log and displays as full-screen text (see Section 7).

6.3 Entry Formatting by Event Category

Event Category	Formatting
COLLECT	Text preceded by a small colored pip matching the petal type color (e.g., #CC0000 for Guilt, #FF6B8A for Desire, #FF0000 for Rage, #D4A574 for

Event Category	Formatting
	Mercy).
HAZARD	Text preceded by a small red thorn icon or pip (#8B0000).
THRESHOLD	Text displayed with a brief full-width underline pulse in amber (#D4A574). Duration: 800ms.
TERMINAL	Bypass the log entirely. Display as full-screen overlay text (see Section 7).

7. Terminal Screen Specification

7.1 Common Elements

All terminal (end-of-game) screens share the following foundational behavior:

- Full-screen overlay fading in over **1.5 seconds**.
- Game world remains visible but darkens beneath the overlay.
- Overlay background: #0A0A0A at 85% opacity.
- All terminal screens include: ending title, narrative text, final state readout, and restart prompt.

7.2 Consumed by Thorns (Lose — EVT-030)

Element	Specification
Title	"CONSUMED BY THORNS" — Crimson (#8B0000), Cambria serif, 28px, centered
Narrative Text	Canonical narrative from EVT-030 — italic, #F0E6D3, 14px, centered beneath title
Visual Effect	Thorn vine borders creep inward from screen edges during the 1.5s fade-in
Final Readout	"Corruption: [value] Redemption: [value] Score: [value]" — #666666, 10px

Element	Specification
Restart Prompt	"Press ENTER to begin again" — #C0C0C0, 12px, pulsing opacity, positioned at bottom

7.3 Mercy Dominant (Win A — EVT-031)

Element	Specification
Title	"MERCY DOMINANT" — Warm gold (#D4A574), Cambria serif, 28px, centered
Subtitle	"The thorns retreat" — #C0C0C0, 14px
Narrative Text	Canonical narrative from EVT-031 — italic, #F0E6D3, 14px, centered
Visual Effect	Golden light radiates outward from center during fade-in; thorn borders recede; rose petal particles drift downward
Final Readout	Same format as lose screen
Restart Prompt	"Press ENTER to walk this garden again" — #D4A574, 12px, pulsing

7.4 Bound but Unbroken (Win B — EVT-032)

Element	Specification
Title	"BOUND BUT UNBROKEN" — mixed crimson-and-gold (left half crimson #8B0000, right half gold #D4A574, or alternating letters), Cambria serif, 28px, centered
Subtitle	"The thorns remain — but they are yours now" — #C0C0C0, 14px
Narrative Text	Canonical narrative from EVT-032 — italic, #F0E6D3, 14px, centered
Visual Effect	Thorns remain at screen edges but glow with subtle gold veins; subdued amber particles; atmosphere is tense

Element	Specification
	but resolved
Final Readout	Same format as lose screen
Restart Prompt	"Press ENTER to walk this garden again" — #C0C0C0, 12px, pulsing

8. Visual Effects Catalog

All transient visual effects, their trigger events, and their governed specifications. Each effect is tied to a canonical event ID — no effect may fire without a corresponding event dispatch.

Effect ID	Name	Trigger Event	Duration	Visual Description	Layer
VFX-001	Guilt Burn	EVT-001 (Collect Guilt)	300ms	Petal shrinks to center while emitting red-black smoke particles; brief red flash at collection point.	Effect
VFX-002	Desire Split	EVT-002 (Collect Desire)	400ms	Petal splits into two fragments — one red, one gold — that drift apart and dissolve; dual-color particle spray.	Effect
VFX-003	Rage Burst	EVT-003 (Collect Rage)	250ms	Petal explodes outward with ember	Effect

Effect ID	Name	Trigger Event	Duration	Visual Description	Layer
				particles and radial shockwave line; screen shakes 2px for 150ms.	
VFX-004	Mercy Halo	EVT-004 (Collect Mercy)	500ms	Golden expanding ring radiates from collection point; soft light motes drift toward player; brief warm screen-edge glow.	Effect
VFX-005	Thorn Pierce	EVT-010 (Thorn Contact, initial)	100ms flash + 500ms fade	Red pulse from contact point; screen edge flashes crimson; player sprite flickers.	Effect
VFX-006	Thorn Linger	EVT-011 (Sustained Thorn, 3s)	800ms	Dark red vein pattern pulses outward from player; screen vignette deepens momentarily.	Effect
VFX-007	Corruption Warning Pulse	EVT-020 (Corruption ≥ 50)	2,000ms	Full-screen crimson border	Overlay

Effect ID	Name	Trigger Event	Duration	Visual Description	Layer
				pulse (fade in, hold, fade out); corruption meter flashes. One-shot.	
VFX-008	Corruption Critical Pulse	EVT-021 (Corruption ≥ 75)	3,000ms	Intense screen-edge vignette in dark violet; player sprite develops visible thorn tendrils; environment darkens permanently until session end or corruption decreases. One-shot.	Overlay + Entity modification
VFX-009	Redemption Stir	EVT-022 (Redemption ≥ 30)	2,000ms	Brief warm golden pulse from player center; single gold particle trail rises. One-shot.	Effect
VFX-010	Altar Awakening	EVT-023 (Redemption ≥ 65)	3,000ms + permanent	Golden light column rises from altar; altar engravings illuminate; warm particles begin	Entity modification

Effect ID	Name	Trigger Event	Duration	Visual Description	Layer
				orbiting altar permanently. One-shot trigger, permanent visual.	
VFX-011	Altar Rejection	EVT-033 (Altar Rejected)	600ms	Crimson barrier flash at altar boundary; player pushed back visually; altar dims momentarily.	Effect
VFX-012	Score Milestone Flash	EVT-024 (Score thresholds)	500ms	Score display briefly enlarges and flashes white; subtle silver particles.	HUD

9. Audio Specification

9.1 Audio Philosophy

Audio in Bloodrose serves the same governed principle as visuals: every sound is state-justified. Ambient soundscape shifts with corruption and redemption. Event sounds are tied to canonical events. There is no random audio — every sound the player hears maps to a state value or event ID defined in this specification. If it plays, it must be traceable. If it isn't traceable, it must not play.

The audio aesthetic is orchestral, tragic, intimate. Low strings and piano for base atmosphere. Dissonant textures for corruption. Warm resonance for redemption.

Percussive impacts for hazard contact. The audio should feel like a score for a gothic romance — not action music, not horror stingers, but the sound of a heart-garden breathing. Lady Isolde's world has a pulse, and that pulse is governed by state.

9.2 Ambient Soundscape

State Range	Soundscape Description	Instruments / Textures	Volume Level
corruption 0–24, redemption any	Quiet garden ambiance — distant wind, faint water, occasional cricket	Soft pad	0.3 (base)
corruption 25–49	Garden ambiance darkens — wind becomes colder, faint low drone added	Cello sustain	0.35
corruption 50–74	Tension layer — dissonant string cluster fades in, heartbeat-like low pulse	Viola tremolo	0.4
corruption 75–99	Critical layer — full dissonant drone, rhythmic heartbeat, cracking/thorny organic sounds	Low brass	0.5
redemption 30–64	Warm undertone added beneath whatever corruption layer is active — distant piano note, single high string	Piano + high string	0.2 (additive)
redemption 65+	Redemption theme rises — gentle piano melody, warm string swell. Coexists with corruption layer if	Piano melody + warm strings	0.3 (additive)

State Range	Soundscape Description	Instruments / Textures	Volume Level
	both active, creating emotional tension.		

9.3 Event Audio

Event / Category	Sound Description	Duration	Volume
Collect Guilt (EVT-001)	Low reverberant impact + exhaled whisper texture	800ms	0.5
Collect Desire (EVT-002)	Ambiguous chime — warm but with dissonant undertone. Glass bell + distant minor chord.	600ms	0.4
Collect Rage (EVT-003)	Sharp percussive strike + fire crackle. Aggressive staccato.	400ms	0.6
Collect Mercy (EVT-004)	Clear bell tone + ascending warm arpeggio. Clean, breathlike.	1,000ms	0.45
Thorn Contact (EVT-010)	Organic scraping/piercing sound. Short, sharp.	200ms	0.5
Thorn Sustained (EVT-011)	Deeper organic groan + heartbeat intensification	1,500ms	0.4
Corruption Warning (EVT-020)	Low brass swell + dissonant cluster	2,000ms	0.5
Corruption Critical (EVT-021)	Intense drone swell + cracking sounds	3,000ms	0.6
Redemption	Single clear piano	1,500ms	0.35

Event / Category	Sound Description	Duration	Volume
Emerging (EVT-022)	note + faint string warmth		
Altar Awakening (EVT-023)	Rising warm chord + resonant bell	3,000ms	0.5
Altar Rejected (EVT-033)	Dull thud + metallic rejection clang	500ms	0.5
Consumed by Thorns (EVT-030)	Crushing organic sound, fading heartbeat, silence	4,000ms (fading to 0)	0.5 → 0
Mercy Dominant (EVT-031)	Swelling warm strings + clear bell resolution + exhale	5,000ms	0.5 → 0.6
Bound but Unbroken (EVT-032)	Tense string sustain + warm note fighting through + final unresolved chord	5,000ms	0.5

10. Input Handling

10.1 Input Capture

- The Render Layer captures keyboard input events (keydown / keyup for WASD and arrow keys).
- Input is translated to movement vectors and forwarded to the Entity Layer.
- The Render Layer does **not** interpret input as game logic — it captures and forwards only.

10.2 Input-to-Movement Mapping

Key	Direction	Vector
W / ArrowUp	Up	(0, −1)
S / ArrowDown	Down	(0, +1)

Key	Direction	Vector
A / ArrowLeft	Left	(−1, 0)
D / ArrowRight	Right	(+1, 0)
Diagonal (e.g., W+D)	Up-Right	Normalized to unit length
ENTER	Restart	Only active on terminal screens

10.3 Input Feel

- Movement must feel immediate — **no acceleration ramp** in v0.1.
- Movement speed: constant, defined in Data Spec (default: 3 units per second).
- No input buffering — current key state is read each tick.
- No mouse input in v0.1.

11. Performance Specification

11.1 Target Performance

Metric	Target
Frame Rate (target)	60 FPS
Frame Rate (minimum)	30 FPS
Frame Budget	16.67ms per frame at 60 FPS

Render Stage Budget Breakdown (approximate, within 16.67ms total):

Stage	Budget
State Read	< 0.5ms
Environment Render	< 3ms
Entity Render	< 3ms
Effect Render	< 2ms

Stage	Budget
HUD Render	< 1ms
Narrative Render	< 1ms
Terminal Render	< 2ms (only when active)
Buffer (overhead / variance)	~4ms

11.2 Rendering Optimization Rules

13. Only redraw entities that have moved or changed state since last frame.
Dirty-rect optimization is encouraged but not required in v0.1.
14. Particle systems must have hard caps: **maximum 50 active particles per effect, maximum 200 total active particles** on screen.
15. Visual effects must auto-cleanup after their defined duration — **no orphaned effects**.
16. Off-screen entities are not rendered (frustum culling at canvas bounds).
17. Audio assets should be preloaded at session start, not streamed during gameplay.

12. Render Governance Rules

NON-NEGOTIABLE — Render Pipeline Governance

The following rules are architecturally enforced and validated by the Drift Detection Harness. Violations are build-breaking. No exceptions.

18. **No State Mutation.** The Render Pipeline must never modify game state.
(Modular Architecture §4.7; Drift Detection DFT-012 — CRITICAL violation.)
19. **Full Traceability.** Every visual element must be traceable to a canonical state value, event, or entity. No decorative elements without state justification in this version.

20. **HUD Convergence.** HUD meters must converge to true state values within 500ms. Visual smoothing is cosmetic only and must not misrepresent state.
21. **Narrative Fidelity.** The narrative log must display exactly what the Event System dispatches — no suppression, fabrication, delay beyond the 500ms fade-in, or editorial modification.
22. **Terminal Text Integrity.** Terminal screens must display the canonical narrative text from the Event Table for the triggered ending — no alternative text.
23. **Audio Justification.** Audio must not play sounds that are not mapped to a state range or event ID in this specification.
24. **Effect Termination.** Visual effects must terminate at their defined duration. No persistent effects unless explicitly marked permanent in Section 8.
25. **Graceful Degradation.** The Render Pipeline must gracefully handle missing assets (placeholder rendering with console warning, not crash).
26. **Palette Governance.** Color palette deviations from the canonical palette table (Section 4.1) require a formal amendment to this document.
27. **No Hardcoded Timing.** All timing values (durations, fade times, cycles) are governed constants sourced from Data Spec — not hardcoded in render code.

13. Implementation Phases

Phase 1 — v0.1 (Prototype)

- Canvas-based rendering in single-file HTML.
- Simple geometric shapes for entities (colored circles/rectangles) — no sprite art required.
- Basic HUD with text-rendered meters (no gradient fills required).
- Narrative log as simple text append with CSS opacity transitions.
- Terminal screens as DOM overlays.
- No audio in Phase 1 (optional stretch goal).

- State-driven color mapping implemented as canvas background color shifts.
- Minimal particle effects (optional).

Phase 2 — Modular

- Render Layer as a dedicated module per Modular Architecture.
- Sprite-based entity rendering with proper art assets.
- Gradient meter fills with threshold markers.
- Full narrative log with fade behavior.
- State-driven environmental effects (tint overlays, vignettes).
- Full visual effects catalog implemented.
- Audio system with ambient layers and event sounds.
- Particle system with governed caps.

Phase 3 — Production

- Polished visual assets (professional sprite art, tilesets, environmental detail).
- Full audio implementation with orchestral/ambient assets.
- Advanced visual effects (shader-based if moving to WebGL).
- Performance profiling and optimization.
- Alternative render targets for cross-media (feeds into Cross-Media Adaptation Spec).

14. Integration Points

The Render Pipeline does not exist in isolation. It connects to the broader Bloodrose artifact system through the following governed interfaces:

Artifact	Relationship
Canonical Event Table (Artifact 1)	Visual effects and audio cues map 1:1 to Event Table entries. Every VFX-ID references an EVT-ID. If an event exists

Artifact	Relationship
	in the table but has no VFX mapping here, it has no visual effect — by design, not omission.
Modular Architecture (Artifact 4)	The Render Layer is Module 4.7 — a pure consumer with defined inputs from State Manager, Entity Layer, and Event System. Module boundaries are enforced; the Render Layer exposes no public write interface.
Drift Detection Harness (Artifact 5)	DFT-012 (Render State Mutation) and DFT-013 (Aesthetic Violation) are enforced against this pipeline. Palette analysis for DFT-013 uses the canonical color palette from Section 4.1.
Story Forge Integration Layer (Artifact 6)	Aesthetic canon locks (hard lock on crimson/black/gothic palette; soft lock on specific asset designs) are enforced through Story Forge's Canon Lock Registry.
Cross-Media Adaptation Spec (Artifact 8, forthcoming)	This pipeline is the game render target. Alternative render targets (visual novel, cinematic) will have their own render specs but share the same color palette and aesthetic direction.
Operator Surface Spec (Artifact 3)	The Operator Surface renders alongside this pipeline but in raw/unglamoured form. Both consume the same state; neither modifies it.

15. Amendment Log

Amendment ID	Date	Description	Author	Approved By
AMD-000	April 21, 2026	Initial Render Pipeline Spec created.	System Architect	Pending Review

*Bloodrose — Render Pipeline Spec v0.1 — Governed Presentation Architecture —
April 21, 2026*

This document is a governed artifact. Modifications require formal amendment.